Mawlana Bhashani Science and Technology University

Department of Information and Communication Technology

4th year 1st Semester B. Sc (Engg.) Final Examination 2025

Course Title: Digital Image Processing Course Code: ICT 4103

Full Marks: 70 Time: 3 hour

Answer any five from the following questions

1. (a) Compute the Haar transform of the following 2 × 2 image. 4

$$\begin{bmatrix} 3 & -1 \\ 6 & 2 \end{bmatrix}$$

(b) Write the JPEG compression algorithm. 6

(c) Find the difference between lossy and lossless compression technique. 4

2. (a) Define: 4-connectivity, 8-connectivity and m-connectivity. 6
Consider the following image:

$$A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

Assume that the set of foreground pixels is $V = \{1\}$

(i) Determine whether the pixels at (1, 2) and (2, 3) are connected under m-connectivity.

(ii) Identify all m-connected components

(iii) Compare the result with 4-connectivity and 8-connectivity.

(b) Analyze the different pixel distance measurement techniques with suitable example. 3

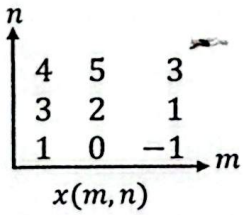
(c) Describe least square filtering technique for image restoration problem formulated as $y = Hx + n$ where x is original image, y is the observed image, n is noise and H is the non-invertible system matrix. 5

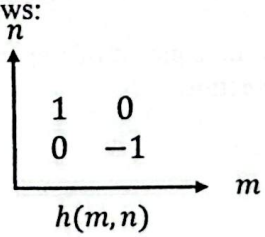
3. (a) State the convolution and correlation theorems. 2

(b) Write short notes on: Contrast stretching, bit plane slicing, and power law (gamma) transformation. 6

(c) Sate the main objectives of image enhancement techniques. 2

(d) Determine the convolution of $x(m,n)$ with $h(m,n)$ where $x(m,n)$ and $h(m,n)$ are given as follows: 4





4. (a) Define the mean square error (MSE) and signal-to-noise ratio in the context of degraded images. 2

(b) Describe the basic model of the image degradation and restoration process. 4

(c) Explain dilation and erosion with example. 6

(d) Write down the applications of digital image processing. 2

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5. (a) Implement nearest neighbor interpolation to enlarge a 4×4 grayscale image to a 5×5 grayscale image. 4

(b) Analyze all the geometric transformation in digital image processing. 4

(c) Explain in detail the effect of varying the number of gray levels in the image. 3

(d) Describe the back propagation network algorithm for image compression technique. 3

6. (a) What is compression ratio? Give the major building blocks of an image compression model. 4

(b) Explain the different types of redundancies normally available in an image. 3

(c) Construct the Huffman codes for the source symbols the probabilities which are tabulated below. Compute the entropy. 7

Symbol	S1	S2	S3	S4	S5	S6	S7
Probabilities	0.19	0.21	0.25	0.08	0.16	0.06	0.02

7. (a) Define histogram. Compute and draw the histogram of the below matrix: 7

$$A = \begin{bmatrix} 9 & 14 & 1 & 8 & 15 \\ 7 & 5 & 7 & 14 & 9 \\ 4 & 6 & 13 & 14 & 9 \\ 10 & 12 & 14 & 7 & 3 \\ 11 & 9 & 14 & 2 & 9 \end{bmatrix}$$

(b) Write short notes on Fourier transform. 3

(c) Write notes on Laplacian of Gaussian filtering technique. 4

8. (a) Derive the CMY transformations to generate the complement of a color image. 4

(b) Compare the arithmetic mean filter and geometric mean filter. 2

(c) Determine the filtered images by applying a 3×3 mean filter to the given image with the pixel values. 4

$$\begin{bmatrix} 20 & 30 & 25 \\ 255 & 40 & 50 \\ 10 & 60 & 35 \end{bmatrix}$$

(d) Discuss the concept of Laplacian of Gaussian (LoG) operator for edge detection. 4

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